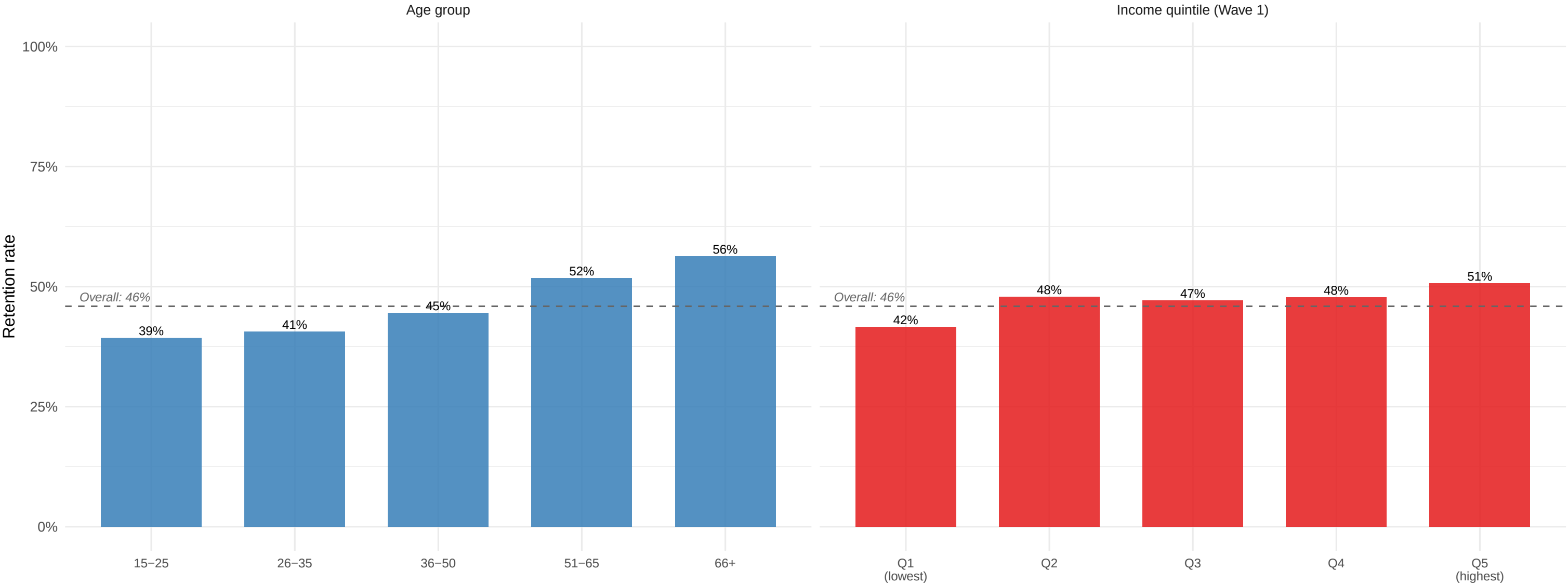


SIPP 2018 Panel: Attrition, Income Dynamics & Design Effects

Survey of Income and Program Participation, 2018 Panel (Waves 1–3, ref years 2017–2019). 63,915 persons in Wave 1 (December). Design: svrepdesign(type='Fay', rho=0.5, mse=TRUE) with 240 replicate weights. First PANEL survey in our analysis series.

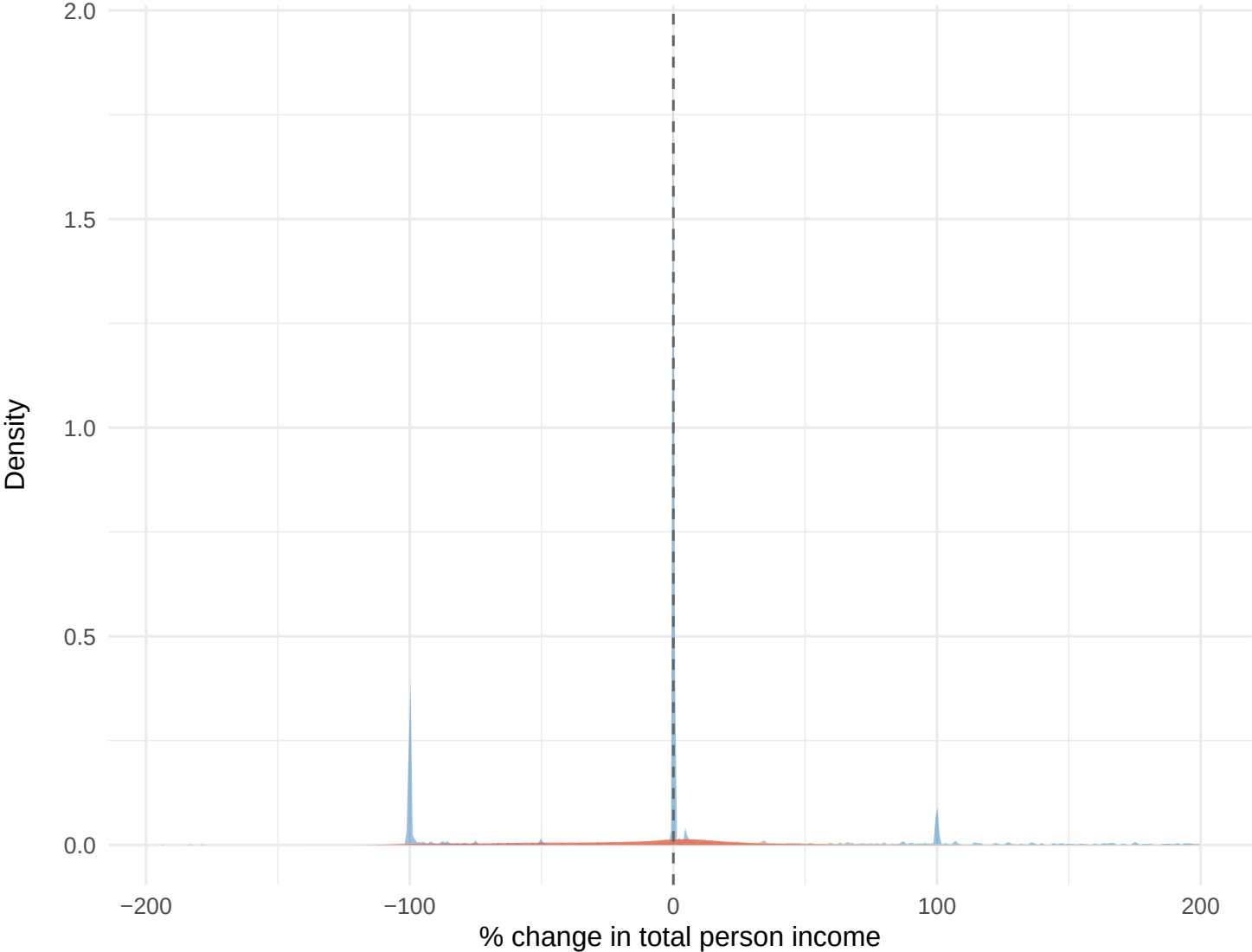
A. Who Drops Out? Retention from Wave 1 to Wave 3

Young adults and low-income respondents have the lowest retention. This is MNAR: the probability of staying depends on income itself. 63,915 Wave 1 respondents tracked.



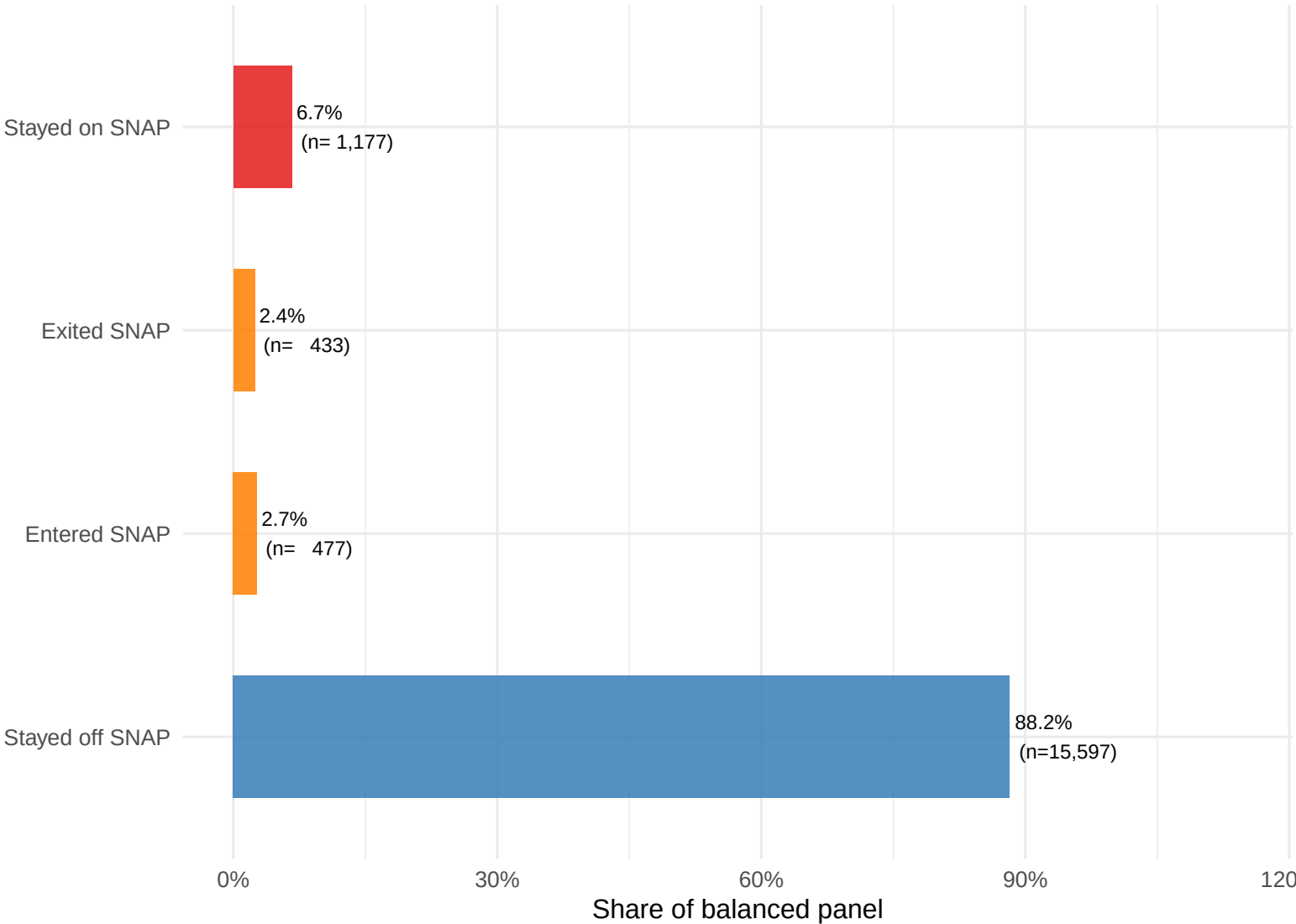
B. Income Change: Wave 1 to Wave 3

Percent change in total person income for balanced panel. Low-income respondents (Q1) show the widest spread. Cross-sectional surveys see none of this within-person volatility.



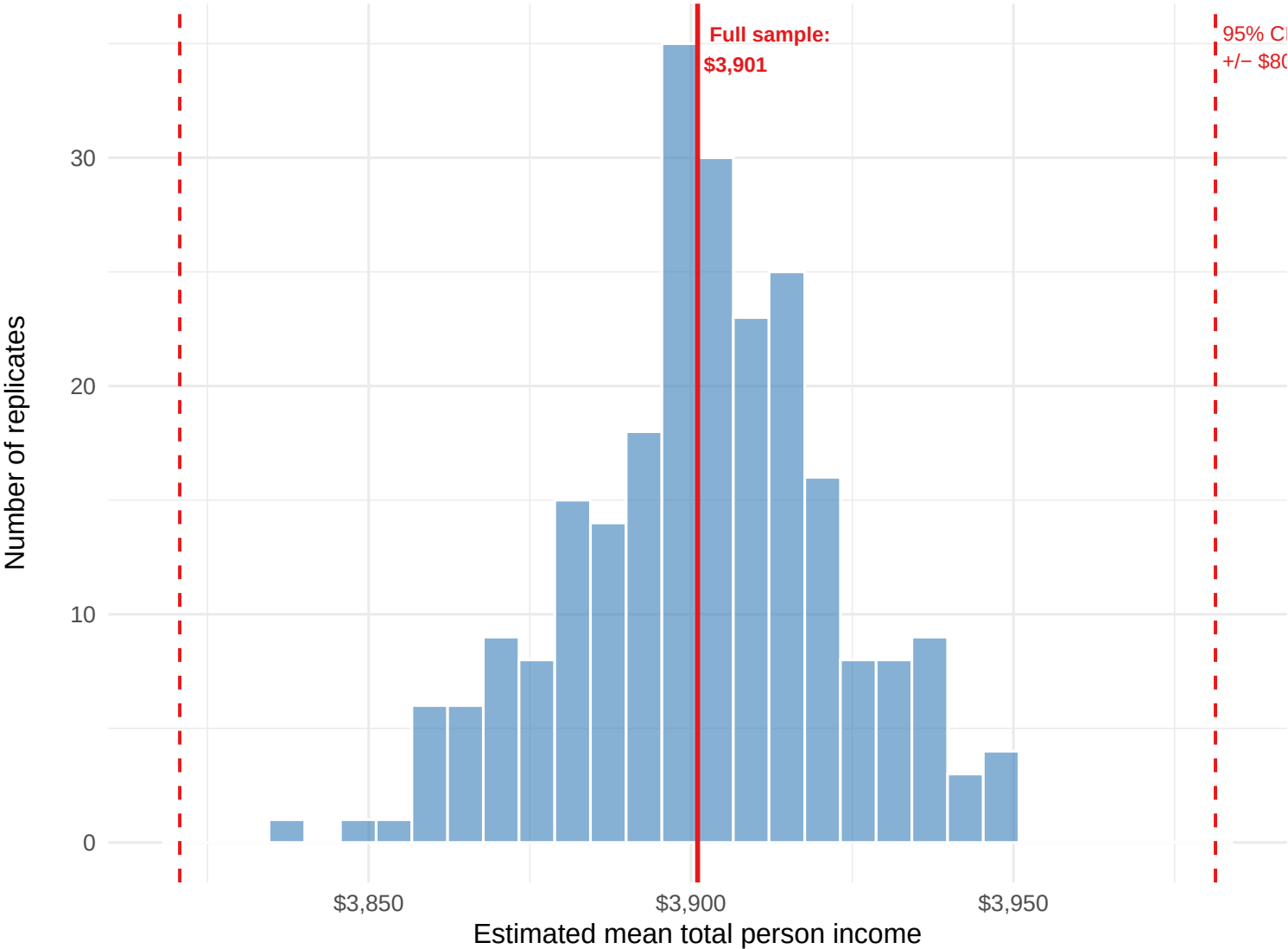
C. SNAP Dynamics: Who Enters and Exits? (Wave 1 to 3)

Panel data reveals program churn invisible to cross-sectional surveys. A snapshot shows the stock; SIPP shows the flow.



D. How Fay's BRR Works: 240 Replicate Estimates of Mean Income

Each bar = a replicate estimate (weights perturbed by rho=0.5). The spread of replicates determines the SE. Unlike Taylor linearization (GSS/NHANES), BRR doesn't need PSU/strata -- the replicates encode the design directly.



E. Weight Dispersion Grows with Attrition

Person weight (WPFINWGT) distribution by wave (top 1% trimmed). Later waves show heavier tails as attrition-adjusted weights upweight remaining respondents.

